

could finish in a few days. Each day, the solution seemed to be just around the corner, and during the fifth week I had a working prototype.

During this entire process, I learned a lot of details about how the graphing library worked and what its capabilities and limitations were. I then spent a few hours writing up these details into a page-long comment that I placed in the source code. I knew that anyone else who needed to make changes to my code later would encounter all the same, seemingly simple issues I had, and that the documentation I was writing would save them literally weeks of effort.

These *lessons-learned* comments, as I call them, might span several paragraphs, making them seem out of place in a source code file. But the information they contain is gold for anyone who needs to maintain this code. Don't be afraid to write lengthy, detailed comments in your source code file to explain how something works. To other programmers, many of these details will be unknown, misunderstood, or easily overlooked. Software developers who don't need them can easily skip past them, but developers who do need them will be grateful for them. Remember that, as with other comments, a lessons-learned comment is not the same as module or function documentation (which docstrings handle). It also isn't a tutorial or how-to guide aimed at users of the software. Instead, lessons-learned comments are for developers reading the source code.

Because my lessons-learned comment concerned an open source graph library and could be useful to others, I took a moment to post it as an answer to the public question-and-answer site <https://stackoverflow.org>, where others in a similar situation could find it.

Legal Comments

Some software companies or open source projects have a policy of including copyright, software license, and authorship information in comments at the top of each source code file for legal reasons. These annotations should consist of a few lines at most and look something like this:

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See license.txt for the full text."""
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